

# Nicholas Curwood

[LinkedIn](#) | [Personal Website](#) | (858) 900-5573 | [Email](#) | [GitHub](#) | Seattle, WA

## SUMMARY

*Versatile and security-conscious Engineer with hands-on experience in full-stack development, cybersecurity, and cloud infrastructure. With industry experience reviewing and securing enterprise software, I bring a unique perspective to building and maintaining scalable and secure applications.*

## EXPERIENCE

### Security Analyst I | Transaction Network Services

Seattle, WA, USA

05/2024 - Present

- Authored the organization's first AI Governance Policy to proactively address emerging security and compliance risks posed by generative AI.
- Reviewed 300+ enterprise-wide software requests, evaluating SaaS and development tools for compliance with internal security policies and potential exposure to third-party risks.
- Aggregated and reconciled user access data from multiple sources to create centralized views of server membership for review; supported access control audits and ensured consistency with NIST 800-53 Rev. 5, ISO 27001, and PCI DSS standards.
- Conducted in-depth malware analysis in a CrowdStrike sandbox environment to identify 50+ communication-based threat vectors and reported my findings through KnowBe4's PhishER solution.
- Maintained risk tracking and incident reports using Salesforce ticketing systems, ensuring timely resolution and cross-team visibility.
- Administered file integrity monitoring via Tripwire to detect unauthorized changes in critical configurations and support compliance audits.
- Launched region-specific phishing simulations through KnowBe4, tracked click-through rates, and used metrics to improve employee security behavior over time.

### Software Development Intern | Computing For All

Seattle, WA, USA

06/2025 - 08/2025

- Collaborated on the design and development of a full-stack web application focused on supporting independent living.
- Coordinated team efforts by reviewing code contributions, resolving team blockers, and guiding task assignments to bring about the harmonious delivery of features.
- Built secure user authentication with login functionality and HTTPS integration, ensuring data protection through TLS.
- Presented the final product to a panel of peers, demonstrating technical implementation and user-centric design decisions.

### Algorithms Grading Assistant | University of Washington

Tacoma, WA, USA

06/2023 - 08/2023

- Collaborated closely with Professor Chris Marriott in the TCSS 343 Algorithms course, grading and offering feedback for class of 35 students
- Demonstrated in-depth expertise in dynamic programming, recurrence relations, and a wide array of algorithms enhancing the academic experience and understanding of students through detailed feedback and actionable insights.

## PROJECTS

### Retrieval-Augmented Generation (RAG) Chatbot | [Link](#)

2025

*Python | JavaScript | Flask | LangChain | ChromaDB | LM Studio | Hugging Face | Git*

- Developed a RAG-based AI chatbot that enhances responses by retrieving vectorized knowledge from a ChromaDB database, integrated LM Studio for local inference, and constructed a friendly JavaScript frontend to facilitate user interaction.
- Organized the chat interface to provides factual responses along with cited sources, ensuring accountability and traceability of information for the user to review references.
- Engineered memory-efficient processing that handles large document collections through streaming and batched operations.

### Portfolio Website | [Link](#)

2024

*JavaScript | HTML | CSS | ReactJS | CodeCommit | AWS | S3 | Route 53 | CloudFront*

- Leveraged JSX to architect a dynamic React website showcasing professional background, educational takeaways, technical skill set, project descriptions, certifications, and a means of direct contact.
- Successfully hosted the site as a static webpage on Amazon S3, integrated with a personalized domain via Route 53, and globally distributed using Amazon CloudFront for optimized performance and delivery.

### Artificial Life Vision Research | [Link](#)

2023

*JavaScript | HTML | Python | SQL | Machine Learning | Git*

- Collaborated in a 4-member team in an artificial life research venture, enhancing a virtual predator's hunting capabilities through binary vision.
- Implemented enhancements using JavaScript and utilized Python for key performance analysis, improving simulation realism.

### Sentiment Analysis Research | [Link](#)

2023

*Python | Keras API | Natural Language Processing | Machine Learning | Git*

- Conducted comparative research on Convolutional Neural Networks (CNN) versus Bag-of-Words (BoW) algorithms for sentiment analysis on X (Twitter) data, focused on the ethics of AI-generated art.
- Leveraged Python and the Keras API, achieving a 20% higher accuracy rate with CNN, demonstrating its superior efficacy in distinguishing sentiment within natural language.

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**Dungeon Adventure Game** | [Link](#)

**2022**

*Java | Git | SQL | Agile | Object Oriented Programming | Model-View-Controller | Full-Stack Development*

- Led a team of 3 to develop an intricate Java-based dungeon crawler game, featuring randomized layouts, diverse player classes, and a variety of enemies and bosses
- Adopted Agile methodologies such as Scrum and Continuous Integration to ensure rapid iteration progress while maintaining adaptability
- Resulted in a user-friendly and engaging game experience that exceeded expectations, coupled with robust technical documentation and evenly maintained code quality through bi-weekly code reviews

## EDUCATION

**Bachelor of Computer Science**

**University of Washington**

**2021 - 2023**

## CERTIFICATIONS

**Security+**

**CompTIA**

**Valid Until:**

**01/2027**

**AWS Cloud Practitioner**

**Amazon Web Services**

**Valid Until:**

**10/2026**

**OCI Cloud Foundations**

**Oracle Cloud Infrastructure**

**Valid Until:**

**3/2026**

## SKILLS

### Security & Compliance:

CrowdStrike | Tripwire | KnowBe4 | File Integrity Monitoring (FIM) | NIST 800-53 Rev.5 | ISO 27001 | PCI DSS | Third-Party SaaS Risk Management | Security Policy Development

### Technical & Scripting:

JavaScript | Python | Java | C# | REST APIs | HTML | CSS | SQL | NoSQL | Git | Object Oriented Programming (OOP) | Dynamic Programming | Machine Learning | Full-Stack Development

### IT Tools:

Amazon Web Services (AWS) | Salesforce | Endpoint Detection & Response (EDR) | Identity & Access Management (IAM)